

# Dong-Kyum Kim

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## Research Interests

- **Interdisciplinary physicist and AI researcher**, guided by physical principles and observations from neuroscience.
- **Knowledge in large language models**: uncovering how models represent, retrieve, update, and forget information through mechanistic interpretability, model editing, and machine unlearning.
- **From physical and biological systems to AI**: drawing on nonequilibrium statistical physics and biological learning and memory to build safer, more reliable, and controllable AI systems.

## Education

**Korea Advanced Institute of Science and Technology (KAIST)** 2016–2022

Ph.D. in Physics

Advisor: Prof. [Hawoong Jeong](#)

Dissertation: *Nonequilibrium Statistical Physics Study using Deep Learning*

**Seoul National University (SNU)**

2011–2015

B.S. in Physics, Minor in Computer Science & Engineering

## Research Experience

**Postdoctoral Researcher**, Max Planck Institute for Security and Privacy (MPI-SP) Oct. 2024–Present

- Hosted by Prof. [Meeyoung Cha](#) (Scientific Director)
- Data Science for Humanity Group

**Postdoctoral Researcher**, Institute for Basic Science (IBS)

Mar. 2022–Sep. 2024

- Hosted by Prof. [Meeyoung Cha](#) (Chief Investigator)
- Data Science Group, Center for Mathematical and Computational Science

**Data Science Intern**, Samsung Electronics

Sep.–Dec. 2017

- Built distributed ML systems with [Daniel Kim](#) (Senior Data Scientist): multi-GPU anomaly image classification (Keras & Spark) and an Elasticsearch-based image search framework

## Publications

<sup>†</sup> denotes equal contribution.

- [1] Jea Kwon<sup>†</sup>, **Dong-Kyum Kim**<sup>†</sup>, Jiwon Kim<sup>†</sup>, Yonghyun Kim, Woong Kook, and Meeyoung Cha. “[AI Engram: In Search of Memory Traces in Artificial Intelligence](#).” *ICML 2026 (Oral)*.
- [2] Minsung Kim, **Dong-Kyum Kim**, Jea Kwon, Nakyeong Yang, Kyomin Jung, and Meeyoung Cha. “[How Training Data Shapes the Use of Parametric and In-Context Knowledge in Language Models](#).” *ACL 2026*.
- [3] **Dong-Kyum Kim**, Minsung Kim, Jea Kwon, Nakyeong Yang, and Meeyoung Cha. “[Bilinear representation mitigates reversal curse and enables consistent model editing](#).” *ICLR 2026*.
- [4] Nakyeong Yang, **Dong-Kyum Kim**, Jea Kwon, Minsung Kim, Kyomin Jung, and Meeyoung Cha. “[Erase or Hide? Suppressing Spurious Unlearning Neurons for Robust Unlearning](#).” *ICLR 2026*.
- [5] Jea Kwon, Jiwon Kim, **Dong-Kyum Kim**, and Meeyoung Cha. “[Moir: Let the Model Direct Its Own Story for Robust Cross-Domain Knowledge Editing](#).” *ICML 2026 Workshop on Mechanistic Interpretability*.
- [6] Minsung Kim, Jea Kwon, **Dong-Kyum Kim**, and Meeyoung Cha. “[In Search of the Engram in LLMs](#).” *ICLR 2025 Blogpost*.
- [7] Gwangsu Kim, **Dong-Kyum Kim**, and Hawoong Jeong. “[Spontaneous emergence of rudimentary music detectors in deep neural networks](#).” *Nature Communications* 2024.

- [8] Jea Kwon, Sunpil Kim, **Dong-Kyum Kim**, Jinhyeong Joo, SoHyung Kim, Meeyoung Cha, and C. Justin Lee. “[SUBTLE: An unsupervised platform with temporal link embedding that maps animal behavior.](#)” *IJCV* (2024).
- [9] **Dong-Kyum Kim**<sup>†</sup>, Jea Kwon<sup>†</sup>, Meeyoung Cha, and C. Justin Lee. “[Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity.](#)” *NeurIPS* 2023.
- [10] Sangyun Lee, **Dong-Kyum Kim**, Jong-Min Park, Won Kyu Kim, Hyunggyu Park, and Jae Sung Lee. “[Multidimensional entropic bound.](#)” *Phys. Rev. Research* **5**, 013194 (2023).
- [11] **Dong-Kyum Kim**<sup>†</sup>, Sangyun Lee<sup>†</sup>, and Hawoong Jeong. “[Estimating entropy production with odd-parity state variables via ML.](#)” *Phys. Rev. Research* **4**, 023051 (2022).
- [12] Youngkyoung Bae, **Dong-Kyum Kim**, and Hawoong Jeong. “[Inferring dissipation maps from videos using CNNs.](#)” *Phys. Rev. Research* **4**, 033094 (2022).
- [13] **Dong-Kyum Kim** and Hawoong Jeong. “[Deep reinforcement learning for feedback control in a collective flashing ratchet.](#)” *Phys. Rev. Research* **3**, L022002 (2021).
- [14] **Dong-Kyum Kim**<sup>†</sup>, Youngkyoung Bae<sup>†</sup>, Sangyun Lee, and Hawoong Jeong. “[Learning Entropy Production via Neural Networks.](#)” *Phys. Rev. Lett.* **125**, 140604 (2020).
- [15] **Dong-Kyum Kim**<sup>†</sup>, Byunghwee Lee<sup>†</sup>, Daniel Kim, and Hawoong Jeong. “[Multi-label classification of historical documents using hierarchical attention.](#)” *J. Korean Phys. Soc.* **76**, 368 (2020).

## Teaching Experience

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|----------------------------------------------------------------------------------------|------------------------|
| <b>Computational Physics Course, KAIST (Daejeon, Korea)</b>                            | May 1, 2023            |
| <i>“Deep learning applications: Nonequilibrium statistical physics study using AI”</i> |                        |
| <b>Computational Physics Course, KAIST (Daejeon, Korea)</b>                            | Apr. 24, 2023          |
| <i>“Deep Learning Introduction”</i>                                                    |                        |
| <b>SNU Physics and AI Winter School (Seoul, Korea)</b>                                 | Feb. 24, 2022          |
| <i>“Exploring Irreversibility via Machine Learning”</i>                                |                        |
| <b>General Physics II, KAIST (Korea)</b>                                               | Fall 2016; Spring 2017 |
| Teaching Assistant                                                                     |                        |

## Talks

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|----------------------------------------------------------------------------------------------------|---------------|
| <b>Spark(l)ing Science, MPI-SP (MB/1 Seminar Room + Zoom)</b>                                      | Jul. 24, 2026 |
| <i>“AI research using agents and notebooks”</i>                                                    |               |
| <b>CREx #23 (Online)</b>                                                                           | Feb. 24, 2026 |
| <i>“Bilinear relational structure fixes reversal curse and enables consistent model editing”</i>   |               |
| <b>IBS Winter School on AI-Boosted Basic Science (Daejeon, Korea)</b>                              | Dec. 13, 2022 |
| <i>“Resemblances between Transformer’s Nonlinearity and NMDA Receptor Dynamics”</i>                |               |
| <b>KIAS CAINS Summer Workshop (Jeju, Korea)</b>                                                    | Sep. 2, 2022  |
| <i>“Working and reference memory in transformers on a navigation task”</i>                         |               |
| <b>KIAS Nonequilibrium Statistical Physics of Complex Systems (Seoul, Korea)</b>                   | Jul. 25, 2022 |
| <i>“Deep reinforcement learning for optimal mechanism in active Brownian particles”</i>            |               |
| <b>KIAS AI and Natural Sciences Seminar (Seoul, Korea)</b>                                         | Apr. 27, 2022 |
| <i>“Nonequilibrium Statistical Physics Study Using Deep Learning”</i>                              |               |
| <b>APCTP Workshop for Physics and Machine Learning (Jeju, Korea)</b>                               | Nov. 26, 2021 |
| <i>“Exploring optimal mechanisms in active Brownian particles via deep reinforcement learning”</i> |               |
| <b>Seoul National University Statistical Physics Seminar (Online)</b>                              | Feb. 1, 2021  |
| <i>“Methods of estimating entropy production”</i>                                                  |               |

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| <b>Korean Physical Society Fall Meeting (Online)</b><br>“Deep reinforcement learning for feedback-controlled flashing ratchets”                          | Nov. 6, 2020  |
| <b>NetSci2020 (Online)</b><br>“Discovering wiring patterns of neural networks via backboning”                                                            | Sep. 22, 2020 |
| <b>Korean Physical Society Spring Meeting (Online)</b><br>“Neural estimator for entropy production”                                                      | Jul. 13, 2020 |
| <b>Quantifying Success Satellite at NetSci2019 (Burlington, Vermont, USA)</b><br>“Quantifying Individual Reputation in Large-scale Historical Documents” | May 27, 2019  |

## Technical Skills

- **Daily driver:** Currently running multiple Claude Code sessions (Max plan, \$200/mo) via tmux as my primary research workflow.
- **ML/Interp:** PyTorch, JAX, TransformerLens, & nnsight.
- **Infrastructure:** Distributed training & inference (Slurm system with hundreds of H100s), large-scale data pipelines

## Academic Service

**Program Committee:** AAAI 2027, CIKM 2026.

**Reviewer:** TMLR, NeurIPS 2026, ICML 2026 Workshop on Mechanistic Interpretability, ICML 2026 (Received Gold Reviewer Award), ICLR 2026, NeurIPS 2025, ICML 2025 Workshop World Models, ICLR 2025 Workshop Re-Align, NeurIPS 2024 Workshop Behavioral ML, ICML 2024 Workshop LLMs and Cognition.

## Awards

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|----------------------------------------------|------|
| <b>Pre-doctoral Fellow of Physics, KAIST</b> | 2021 |
|----------------------------------------------|------|

## In the Press

- **Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity** (*NeurIPS* 2023): [IBS Research News \(English\)](#), [IBS Research News \(Korean\)](#), [Donga Science](#), and [EurekAlert!](#).
- **Learning Entropy Production via Neural Networks** (*Phys. Rev. Lett.* **125**, 140604, 2020): [Physics and High Technology](#).

## References

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